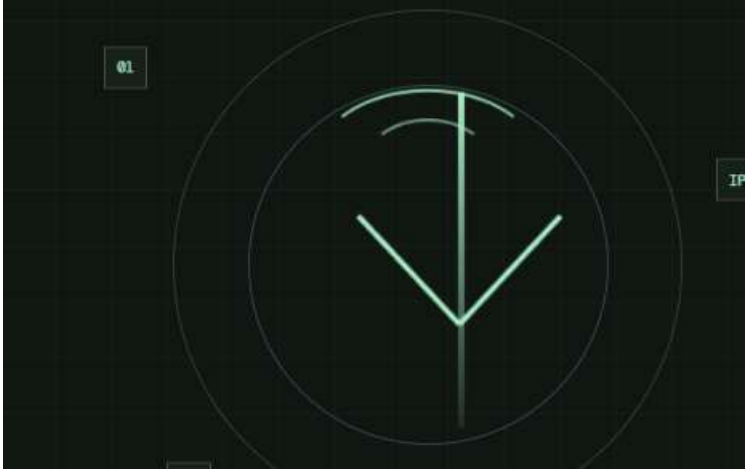
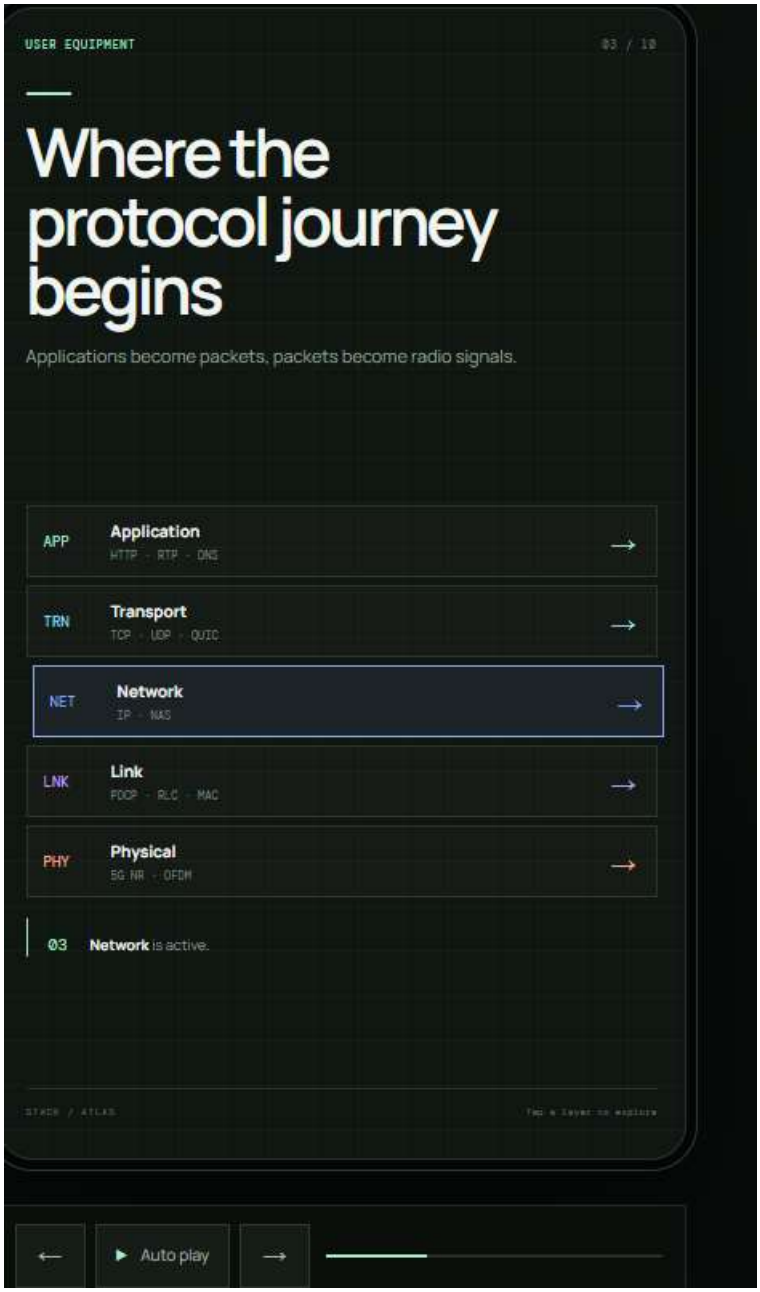
MOBILE SYSTEMS · FIELD GUIDE

Inside the mobile network stack

A visual journey from radio waves to the apps in your hand.




THE BIG PICTURE
02 / 10
One message, many systems
Every tap crosses the device, radio network, mobile core, and internet.
01
UE
Your device
02
gNB
Radio access
03
5GC
Mobile core
04
EDGE
Low latency
05
CLOUD
Internet services
STACK / ATLAS
Swipe to navigate
←
▶ Auto play
→


USER EQUIPMENT
03 / 10
Where the protocol journey begins
Applications become packets, packets become radio signals.
APP
Application
HTTP · RTP · DNS
TRN
Transport
TCP · UDP · QUIC
NET
Network
IP · NAS
LNK
Link
PDCP · RLC · MAC
PHY
Physical
5G NR · OFDM
03 Network is active.
STACK / ATLAS
Tap a layer to explore
Auto play

RADIO ACCESS NETWORK

The air interface

A layered system keeps bits reliable while the user moves.



RRC	Connection & mobility control
PDCP	Ciphering · header compression
RLC	Segmentation · retransmission
MAC	Scheduling · multiplexing
PHY	Modulation · beamforming

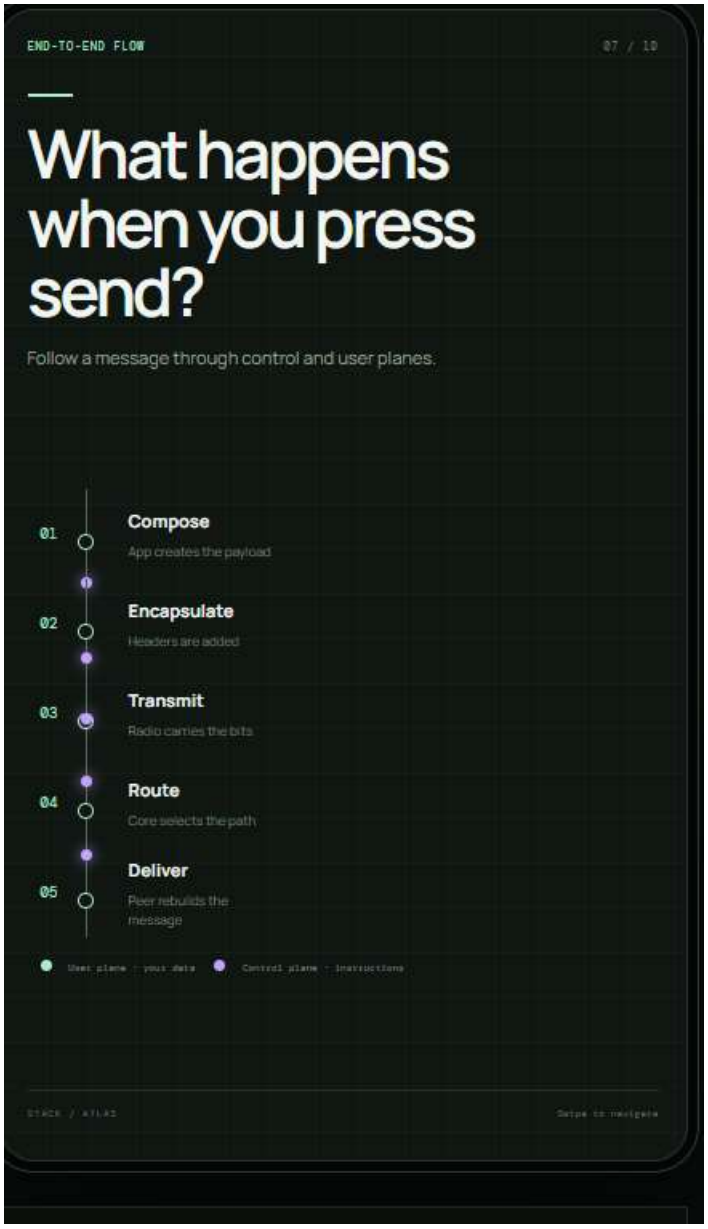
STACK / ATLAS

Swipe to navigate

← ▶ Auto play →


5G CORE
05 / 10
A service-based brain
Specialized network functions authenticate, route, and manage every session.
01
AMF
Access & mobility
02
SMF
Session control
03
UPF
Packet routing
04
UDM
Subscriber data
05
AUSF
Authentication
STACK / ATLAS
Swipe to navigate


EVOLUTION
06 / 10
From 2G voice to 5G services
Each generation reshaped the stack around a new primary experience.
2G
1991
3G
2001
4G
2009
5G
2019
5G
PEAK SPEED
10 Gbps
DESIGNED FOR
Connected everything
NR
5GC
SBA
STACK / ATLAS
Tap a generation to compare


END-TO-END FLOW
07 / 10
What happens when you press send?
Follow a message through control and user planes.
01
Compose
App creates the payload
02
Encapsulate
Headers are added
03
Transmit
Radio carries the bits
04
Route
Core selects the path
05
Deliver
Peer rebuilds the message
User plane · your data
Control plane · instructions
STACK / atlas
Swipe to navigate


TRUST ARCHITECTURE
08 / 10
Security at every layer
Identity, encryption, and integrity controls form overlapping defenses.
01
Identity
SIM / SUPI
02
Authentication
AKA challenge
Trusted session
Defense in depth
03
Encryption
Air interface
04
Integrity
Tamper detection
STACK / ATLAS
Swipe to navigate

The stack is becoming programmable

Cloud-native cores, edge computing, and network slicing point toward 6G.

01 **SLICE**

One network, many virtual lanes



02 **EDGE**

Compute moves closer to people



03 **OPEN RAN**

Interoperable radio components



04 **AI-NATIVE**

Networks learn and adapt



A coordinated conversation

Every layer has one job—and every job depends on the layers around it.



01 Layers simplify complexity.

02 Planes separate data from decisions.

03 Standards make global mobility possible.